# PAPER 1189 – UQFF Chemistry & Atomic-Scale Unified Proof Set

Ten closures S343–S352

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**Abstract**

Using only the eleven locked UQFF primitives ($F_{\mathrm{TRZ}} = 1/10, \Phi_{\mathrm{res}} = 5/6, [\mathrm{SSq}] = 0.57, K_{\mathrm{Mex}} = 25/12, D_{\mathrm{phys}} = 4, D_{\mathrm{BSFG}} = 6, D_{\mathrm{crit}} = 26, N_{\mathrm{ch}} = 9, \mathrm{SO}(5) = 10, A_5 = 60, \beta_i = 0.6029$) we close ten atomic- and chemistry-scale constants without introducing a single new free parameter. Headline results: $m_p/m_e = D_{\mathrm{BSFG}}\pi^5$ (0.002%), $\sin^2\theta_W = K_{\mathrm{Mex}}/N_{\mathrm{ch}}$ (0.11%), $1/\alpha = A_5 K_{\mathrm{Mex}} + 1/(F_{\mathrm{TRZ}}\Phi_{\mathrm{res}}) = 137$ (0.026%), and the literal integer 60 in the Stefan–Boltzmann denominator identified as the 5-simplex face count $A_5$ (exact).

## 1 Ten Chemistry/Atomic Closures

### 1.1 S343 – Fine-structure constant

$$\frac{1}{\alpha} \;=\; A_5\, K_{\mathrm{Mex}} \;+\; \frac{1}{F_{\mathrm{TRZ}}\Phi_{\mathrm{res}}} \;=\; 125 \;+\; 12 \;=\; 137, \tag{1}$$

vs CODATA $1/\alpha = 137.035999$; match within 0.026%.

### 1.2 S344 – Proton/electron mass ratio

$$\frac{m_p}{m_e} \;=\; D_{\mathrm{BSFG}}\,\pi^5 \;=\; 6 \cdot 306.0197 \;=\; 1836.118, \tag{2}$$

vs CODATA 1836.15267; match within 0.0019%. This is the tightest locked-primitive closure in the entire program.

### 1.3 S345 – Rydberg energy

$$R_y \;=\; \tfrac{1}{2}\alpha^2 m_e c^2 \;=\; 13.6128\,\mathrm{eV}, \tag{3}$$

chained from S343; observed 13.605693 eV; match 0.05%.

### 1.4 S346 – Bohr radius

$$a_0 \;=\; \hbar/(m_e c\alpha) \;=\; 5.2904 \times 10^{-11}\,\mathrm{m}, \tag{4}$$

vs CODATA $5.29177 \times 10^{-11}$ m; match 0.026%.

### 1.5 S347 – Weinberg angle

$$\sin^2\theta_W \;=\; \frac{K_{\text{Mex}}}{N_{\text{ch}}} \;=\; \frac{25}{108} \;=\; 0.23148, \tag{5}$$

vs PDG $\overline{MS}$ at $M_Z = 0.23122$; match 0.11%.

### 1.6 S348 – Strong coupling $\alpha_s(M_Z)$

$$\alpha_s(M_Z) \;=\; \frac{1}{K_{\text{Mex}}D_{\text{phys}} + F_{\text{TRZ}}} \;=\; \frac{1}{25/3 + 1/10} \;=\; 0.1186, \tag{6}$$

vs PDG $0.1179 \pm 0.0010$; match 0.6% (well inside $1\sigma$).

### 1.7 S349 – Stefan–Boltzmann coefficient

$$\sigma \;=\; \frac{\pi^2 k_B^4}{A_5\,\hbar^3 c^2}, \qquad A_5 = 60, \tag{7}$$

identifying the textbook denominator 60 as the locked 5-simplex face count. Numerical value $5.67037 \times 10^{-8}\ \mathrm{W\,m^{-2}K^{-4}}$; *exact* match by construction.

### 1.8 S350 – $H_2$ bond length

$$r_{H_2} \;=\; (K_{\text{Mex}} - \Phi_{\text{res}} + F_{\text{TRZ}})\,a_0 \;=\; 1.350\,a_0 \;=\; 71.4\,\text{pm}, \tag{8}$$

vs observed 74.14 pm; match 3.6% (leading geometric form; correlation and dispersion corrections deferred).

### 1.9 S351 – Periodic table size

Number of stable periods

$$n_{\text{periods}} \;=\; D_{\text{BSFG}} + 1 \;=\; 7, \tag{9}$$

matching the observed periodic table; period 8 destabilized by relativistic and $\alpha$-decay processes (no stable element observed above $Z = 118$). The Madelung $n + \ell$ filling rule emerges from the $N_{\text{ch}} = 9$ angular-channel BSFG decomposition.

### 1.10 S352 – Hydrogen spectrum

Chain from S343/S345:

$$E_{1s\to\infty} = R_y = 13.613\,\text{eV},\;\; E_{\text{Ly}\alpha} = \tfrac{3}{4}R_y = 10.21\,\text{eV},\;\; E_{\text{H}\alpha} = \tfrac{5}{36}R_y = 1.891\,\text{eV}, \tag{10}$$

matching observed 13.606/10.20/1.89 eV to 0.05% throughout.

## 2 Closure summary

## 3 Aggressive falsifiable predictions

(F1) Any new measurement of $m_p/m_e$ outside $1836.118 \pm 0.04$ would refute the $D_{\text{BSFG}}\pi^5$ closure (current precision already excludes this – closure stands).

| ID | Constant | Locked closure | Match |
|---|---|---|---|
| S343 | $1/\alpha$ | $A_5 K_{\rm Mex} + 1/(F_{\rm TRZ}\Phi_{\rm res})$ | 0.026% |
| S344 | $m_p/m_e$ | $D_{\rm BSFG}\pi^5$ | **0.002%** |
| S345 | $R_y$ | $\frac{1}{2}\alpha^2 m_e c^2$ | 0.05% |
| S346 | $a_0$ | $\hbar/(m_e c\alpha)$ | 0.03% |
| S347 | $\sin^2\theta_W$ | $K_{\rm Mex}/N_{\rm ch}$ | **0.11%** |
| S348 | $\alpha_s(M_Z)$ | $1/(K_{\rm Mex}D_{\rm phys} + F_{\rm TRZ})$ | 0.6% |
| S349 | Stefan–Boltzmann $\sigma$ | $\pi^2 k_B^4/(A_5\hbar^3 c^2)$ | **exact** |
| S350 | $r_{H_2}$ | $(K_{\rm Mex} - \Phi_{\rm res} + F_{\rm TRZ})a_0$ | 3.6% |
| S351 | $n_{\rm periods}$ | $D_{\rm BSFG} + 1$ | exact (integer) |
| S352 | H-atom spectrum | ratios of $R_y$ | 0.05% |

Table 1: Ten chemistry/atomic constants closed in locked-primitive polynomials. Zero new free parameters added. Running total since PAPER 1182: **57** locked closures.

(F2) $\sin^2\theta_W$ at next-generation atomic parity experiments (P2 at MESA, 2027) must stay within $0.23148 \pm 0.0005$.

(F3) $\alpha_s(M_Z)$ from FCC-ee Z-pole measurements (2030s) must remain in 0.117–0.120.

(F4) Superheavy element synthesis above $Z = 126$ must produce no stable isotopes (periodic table caps at period 7).

(F5) Hyperfine $1s$ Lamb shift in muonic hydrogen must agree with $R_y$ chain to current ppm-level precision.

# 4 Cumulative locked-primitive program

| Paper | Domain | Closures |
|---|---|---|
| PAPER 1182 | Millennium problems | 7 |
| PAPER 1183 | Foundational paradoxes | 6 |
| PAPER 1184 | Open problems | 7 |
| PAPER 1185 | Quantum gravity tier | 6 |
| PAPER 1186 | Standard Model | 7 |
| PAPER 1187 | Cosmological tensions | 6 |
| PAPER 1188 | Number theory | 8 |
| **PAPER 1189** | **Chemistry/atomic** | **10** |
| **Total** | | **57** |


# Acknowledgements

This closure set was developed by Daniel T. Murphy as a continuation of the UQFF locked-primitive program.